Advanced Optimization

Course Project – Autumn 2026

Course instructor: Prof. Hamid Reza Shaker

Project format	Deadline	Submission
Groups of 3–5 Individual work permitted	**21 December 2026** Monday, 23:59 CET	**Digital Exam** PDF report and supporting files

Purpose and expectations

The project gives you an opportunity to apply advanced optimization concepts and computational methods to a realistic problem in energy systems or infrastructure. Select **one** of the two project options below. Your work should demonstrate a sound mathematical formulation, a justified solution approach, numerical experimentation, and critical interpretation of the results.

State assumptions clearly and explain your modelling decisions. Report both successful and unsuccessful approaches where relevant; careful analysis of why something did not work is often as informative as the final solution.

Option A: Optimal EV charging and discharging

Charging electric vehicles (EVs) places new demands on electricity networks, while flexible charging and, where appropriate, vehicle-to-grid operation offer opportunities to manage charging costs and network peaks. In this option, you will examine a published optimization model, reproduce numerical results, and develop a meaningful extension.

Reference paper. Y. He, B. Venkatesh, and L. Guan, "Optimal scheduling for charging and discharging of electric vehicles," *IEEE Transactions on Smart Grid*, vol. 3, no. 3, pp. 1095–1105, 2012.

- **A1.** **Understand the reference model.** Read the paper and prepare a critical summary (**maximum one page**). Explain the objectives, decision variables, principal constraints, assumptions, and optimization method.
- **A2.** **Reproduce the implementation.** Study and run the CVX code provided with the course materials. Relate the code to the mathematical formulation and reproduce at least one numerical result. Document discrepancies or technical difficulties.
- **A3.** **Perform a sensitivity study.** Vary at least **two** relevant parameters (e.g., electricity prices, charging limits, battery capacity, or EV availability). Present and interpret the changes in the schedules and objective value.
- **A4.** **Develop a model extension.** Introduce a meaningful change, such as feeder-capacity limits, renewable-energy coordination, a new tariff design, or user-preference constraints. Formulate the revised objective and constraints.
- **A5.** **Solve and evaluate the extension.** Implement an appropriate optimization method, compare the extended and original models, and discuss feasibility, trade-offs, limitations, and further improvements.

Option B: Your own optimization problem

Identify a relevant, manageable optimization problem in energy infrastructure or energy applications. Possible areas include electricity distribution, district heating, storage, building operation, EV charging, renewable-energy integration, and maintenance planning. You may use real data or clearly documented synthetic data.

- **B1.** **Define the problem.** Explain its practical context and relevance, the decision to be optimized, the problem scope, and the input data.
- **B2.** **Formulate the mathematical model.** Specify decision variables, the objective function, constraints, parameters, units, and assumptions. Make the connection between the application and the formulation explicit.
- **B3.** **Analyse convexity.** Determine whether the formulated problem is convex and justify your answer by considering both the objective and feasible set. If nonconvex, identify the reasons.
- **B4.** **Select and implement a solution method.** Use **CVX** for a convex model. For a nonconvex model, either justify and solve an appropriate convex approximation/reformulation using CVX or apply **particle swarm optimization (PSO)**. Explain how feasibility and constraints are handled.
- **B5.** **Evaluate the solution.** Compare your results with a meaningful baseline or alternative scenario. Study at least **two** relevant parameters, and discuss feasibility, solution quality, and practical implications.
- **B6.** **Reflect and propose improvements.** Identify modelling and numerical limitations and suggest realistic extensions or future improvements.

Report structure and reproducibility

Submit a self-contained report, written clearly enough for the reader to understand and assess the problem and your approach. The following structure is recommended:

1. **Introduction and problem definition:** motivation, objectives, and scope.
2. **Background and related work:** relevant context and literature; for Option A, include the required one-page article summary.
3. **Mathematical formulation:** variables, objective, constraints, and assumptions.
4. **Solution method and implementation:** algorithm, software, parameters, and relevant settings.
5. **Numerical experiments and results:** data, baseline, parameter studies, and clearly labelled figures and tables.
6. **Discussion and conclusions:** interpretation, limitations, lessons learned, and possible extensions.

Include **references** and, where useful, appendices. Provide the relevant MATLAB/CVX scripts or other code, together with sufficient instructions to reproduce the main results.

Submission instructions

Deadline	**Monday, 21 December 2026, at 23:59 (CET).**
Platform	Submit through the university's **Digital Exam** platform. Follow the relevant group-submission instructions displayed there.
Files	Submit the **project report as a PDF**, together with supporting code and reproducibility files in the format permitted by Digital Exam.
Group members	Include the **full names of all group members** on the first page of the report. Groups of **3–5 students** are recommended; individual work is permitted.